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Serum metabolomic profiling reveals potential biomarkers in systemic sclerosis

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Abstract

Background Systemic sclerosis (SSc) is a chronic and systemic autoimmune disease marked by the skin and visceral fibrosis. Metabolic alterations have been found in SSc patients; however, serum metabolomic profiling has not been thoroughly conducted. Our study aimed to identify alterations in the metabolic profile in both SSc patients before and during treatment, as well as in mouse models of fibrosis. Furthermore, the associations between metabolites and clinical parameters and disease progression were explored.

Methods High-performance liquid chromatography quadrupole time-of-flight mass spectrometry (HPLC-Q-TOF-MS)/MS was performed in the serum of 326 human samples and 33 mouse samples. Human samples were collected from 142 healthy controls (HC), 127 newly diagnosed SSc patients without treatment (SSc baseline), and 57 treated SSc patients (SSc treatment). Mouse serum samples were collected from 11 control mice (NaCl), 11 mice with bleomycin (BLM)-induced fibrosis and 11 mice with hypochlorous acid (HOCl)-induced fibrosis. Both univariate analysis and multivariate analysis (orthogonal partial least-squares discriminate analysis (OPLS-DA)) were conducted to unravel differently expressed metabolites. KEGG pathway enrichment analysis was performed to characterize the dysregulated metabolic

pathways in SSc. Associations between metabolites and clinical parameters of SSc patients were identified by Pearson's or Spearman's correlation analysis. Machine learning (ML) algorithms were applied to identify the important metabolites that have the potential to predict the progression of skin fibrosis.

Results The newly diagnosed SSc patients without treatment showed a unique serum metabolic profile compared to HC. Treatment partially corrected the metabolic changes in SSc. Some metabolites (phloretin 2'-O-glucuronide, retinoyl b-glucuronide, all-trans-retinoic acid, and betaine) and metabolic pathways (starch and sucrose metabolism, proline metabolism, androgen and estrogen metabolism, and tryptophan metabolism) were dysregulated in new-onset SSc, but restored upon treatment. Some metabolic changes were associated with treatment response in SSc patients. Metabolic changes observed in SSc patients were mimicked in murine models of SSc, indicating that they may reflect general metabolic changes associated with fibrotic tissue remodeling. Several metabolic changes were associated with SSc clinical parameters. The levels of allysine and all-trans-retinoic acid were negatively correlated, while D-glucuronic acid and hexanoyl carnitine were positively correlated with modified Rodnan skin score (mRSS). In addition, a panel of metabolites including proline betaine, phloretin 2'-O-glucuronide, gamma-linolenic acid and L-cystathionine were associated with the presence of interstitial lung disease (ILD) in SSc. Specific metabolites identified by ML algorithms, such as medicagenic acid 3-O-b-D-glucuronide, 4'-O-methyl-(-)-epicatechin-3'-O-beta-glucuronide, valproic acid glucuronide, have the potential to predict the progression of skin fibrosis.

Conclusions Serum of SSc patients demonstrates profound metabolic changes. Treatment partially restored the metabolic changes in SSc. Moreover, certain metabolic changes were associated with clinical manifestations such as skin fibrosis and ILD, and could predict the progression of skin fibrosis.

Keywords Systemic sclerosis, Metabolomics, Metabolic reprogramming, Machine learning algorithm, Biomarkers

Introduction

Systemic sclerosis (SSc) is a highly heterogeneous autoimmune disease, characterized by vasculopathy, immune disorder, skin and visceral fibrosis[1]. Multiple system involvement can occur in patients with SSc, including interstitial lung disease (ILD), pulmonary arterial hypertension (PAH), myocardial fibrosis, esophageal dysmotility, and renal crisis. Despite the low incidence (10-500/1,000,000 per year), the mortality of SSc is relatively high and the quality of life of patients is greatly impacted[2, 3].

The pathologic mechanisms of SSc have not been fully clarified. Environmental factors acting on genetically predisposed individuals can trigger inflammation, autoimmunity, and fibrosis in SSc[4]. Multiple cell types, including immune cells, fibroblasts, endothelial cells and the inflammatory mediators they produce, participate in the development and progression of SSc[5-

[8]. Several essential signaling pathways are involved in SSc fibrosis, including TGF-β signaling, WNT signaling, JAK/STAT signaling, and Notch signaling [9-12].

Recently, metabolic perturbations have also been demonstrated to play important roles in the pathogenesis of SSc[13-15]. Enhanced glycolysis in fibroblasts is the most significant metabolic change found thus far in SSc. Inhibitors of glycolysis and glutamine metabolism suppressed TGFβ-induced fibrosis in SSc[16]. Other metabolic alterations such as decreased nicotinamide adenine dinucleotide and dyslipidemia, have also been mentioned in SSc.

Metabolomics, which involves uncovering the complete set of metabolites and metabolic pathways, has become a research hotspot in diverse diseases[17, 18]. Through the integration of analysis of the fecal microbiota and plasma metabolites, interactions were found between *Desulfovibrio* and benzene derivatives and glycerophospholipids in SSc patients[19]. L-tyrosine and L-tryptophan may be associated with ILD (36 SSc patients)[20]; increased acetate, lactate and decreased creatine, glucose, histidine were found in SSc patients with PAH (18 SSc patients) [21]. Metabolites such as acetate, fructose, and glutamate were found to be distinct between diffuse SSc (dcSSc) and limited SSc(lcSSc)[22]. However, due to the high heterogeneity of SSc, metabolomics studies with large sample sizes are still needed. No comprehensive metabolomics study has been performed in a longitudinal cohort of new-onset and treated SSc patients. In addition, there is no study on the coanalysis of metabolomics in SSc patients and mouse models.

Here, we performed the largest metabolic study in SSc so far, with unbiased serum metabolic profiling in 184 SSc patients (before and after treatment) and 142 healthy controls (HC), as well as in 33 mice with experimental fibrosis. We revealed the commonly altered metabolites and pathways in both SSc patients and mouse models. By exploring the relationships between serum metabolic profiles and SSc clinical parameters, we identified some metabolites and pathways that may be involved in the pathogenesis of SSc, and have the potential to be biomarkers for SSc.

Materials and methods

Patients

A total of 184 serum samples from SSc patients and 142 serum samples from age- and sex-matched HC were collected at Xiangya Hospital. Among the SSc samples, there were 127 baseline samples from newly diagnosed SSc patients (SSc baseline) and 57 samples from treated SSc patients (SSc treatment). In the SSc treatment group, serum samples were collected from follow-up patients who were part of the 127 newly diagnosed patients. These samples were collected at various intervals (1, 3, 6, 12 or 24 months) after beginning treatment. All enrolled patients fulfilled the 2013 ACR/EULAR criteria for SSc[23] and patients with tumors, infections, other autoimmune diseases, and metabolic diseases were excluded. Improvement in mRSS was defined as a decrease in mRSS by ≥20% from baseline, and these patients were

classified as the skin improved group[24, 25]. Progression in mRSS was considered to have occurred when there was an increase in mRSS by ≥20% from baseline and these patients were classified as the skin progressive group. Stable in mRSS was identified as a change in mRSS by <20% from baseline, and these patients were classified as the skin stable group. The demographic and clinical information of the HC and SSc patients was collected (Supplementary Table 1). Treatment information of the 57 follow-up patients was summarized in Supplementary Table 2. The human sample study was approved by the ethics committee of Xiangya Hospital of Central South University (Changsha, Hunan, China, ID reference number: 201303293). All enrolled patients and HC signed an informed consent form before serum sample collection.

Mouse models

Serum samples were collected from 11 control mice, 11 mice with bleomycin (BLM)-induced fibrosis and 11 mice with hypochlorous acid (HOCl)-induced fibrosis. For BLM-induced fibrosis, 6-week-old female BALB/c mice were subcutaneously administrated the BLM (0.9% NaCl solution) on the upper back every other day for 6 weeks[26]. For the HOCl-induced fibrosis, 6-week-old female BALB/c mice were subcutaneously administrated the freshly prepared HOCL solution every day for 6 weeks as previously described[27, 28]. Control mice were subcutaneously injected with equal volumes of 0.9% NaCl. BALB/c mice were purchased from Shanghai Model Organisms. All animal experimental protocols were approved by the Institutional Animal Care and Use Committees of Central South University.

Sample collection and preparation

Morning fasting blood was collected from HC and SSc patients. Serum samples were separated by centrifugation at 3,000 rpm at 4°C for 10 min. For mouse models, 500ul blood was obtained by eyeball enucleation and was centrifuged at 2,000 rpm at 4°C for 20 min. Serum samples from both humans and mice were collected and immediately stored at -80°C. Before the experiment, serum samples (100ul) were thawed on ice, then 10ul of internal standard (L-2-chlorophenylalanine, 0.3 mg/ml in methanol) was added to each sample. After vortexing for 1 min, 300ul of protein precipitant (methanol/acetonitrile, v/v, 2:1) was added. The mixture was vortexed again for 1 min. Then the mixture was ultrasonicated for 10 min in an ice-water bath and cooled at -20°C for 30 min. After centrifugation at 13,000 rpm at 4°C for 15 min, 200ul of supernatant was collected and stored at -80°C prior to liquid chromatography–mass spectrometry (LC–MS) analysis. Quality control (QC) samples were prepared by mixing equal volumes of all serum samples.

High-performance liquid chromatography quadrupole time-of-flight mass spectrometry (HPLC-Q-TOF-MS)/MS

Untargeted metabolomics analysis of serum samples was performed by HPLC-Q-TOF-MS/MS (Bruker Corporation, USA). Metabolite separation and detection were conducted with ACCLAINMTMRSLC120-C18 columns (100×2.1 mm, 2.2 µm) (Thermo Fisher Scientific, USA) at a

flow rate of 0.4 mL/min at 40°C with a 3 µl injection volume. Mobile phase A was 0.1% formic acid and mobile phase B was 0.1% acetonitrile containing 0.1% formic acid. The gradient flow was set as follows: 2% B for 0-2 min, 50% B for 2-12 min, 90% B for 10-30 min, and then 98% B for 30-60 min. The electrospray ionization (ESI) source was operated in positive (4,500 V) and negative (-4,500 V) ion modes with a mass scanning range of 20-1,000 m/z and the drying temperature was set at 200°C. QC samples were analyzed 10 times at the beginning and after every 10 injections of the random array of samples.

Untargeted metabolomics analyses

Data cleaning, including denoising, peak extraction and alignment, standardization and export were executed by Metaboscape 3.0 software as previously described[29]. The preprocessed data were annotated by the HMDB database (http://www.hmdb.ca/) and SmartFormular database. Then a three-dimensional data matrix containing the peak relative intensities, peak retention time (RT) and mass-to-nucleus ratio (m/z) was obtained for further analysis.

Machine learning (ML) model construction and specific metabolite identification

To evaluate the potential of metabolites to predict the progression of skin fibrosis in SSc patients, 10 ML classifiers were performed in this study: linear support vector machine (SVM), radial basis function SVM (RBF SVM), random forest, nearest neighbors, Gaussian process, decision tree, neural network, adaptive boosting (AdaBoost), Gaussian naive Bayes and quadratic discriminant analysis (QDA). Differential metabolites (FC >1.2 or <0.83, p<0.05 and VIP>1.0) in new-onset SSc patients were used in the construction of ML models. 10 different ML algorithms were compared by the model accuracy scores (mean and 95% confidential interval) and the random forest was the optimal performance classifier. To evaluate the importance of each metabolite in the classification of mRSS progression, the recursive feature elimination (RFE) algorithm was performed, which can efficiently select the most important metabolites in predicting the progression of skin fibrosis in SSc patients. All steps were performed by Python V.3.9.1, scikit-learn V.0.24.2, NumPy V.1.19.5 and pandas V.1.2.1. as previously described[29]. Linear discriminant analysis (LDA), a supervised machine learning algorithm widely used for dimensionality reduction and separation of multiple classes, was also applied to visualize the clustering of SSc patients.

Statistical analysis

Raw data were first normalized by log2 transformation and Pareto scaling. Univariate analysis and multivariate analysis were conducted by MetaboAnalyst 5.0 (https://www.metaboanalyst.ca/) and SIMCA 14.0 software (Umetrics, Umeå, Sweden). In univariate analysis, the fold change (FC) and P value were obtained for each metabolite. For multivariate analysis, orthogonal partial least-squares discriminate analysis (OPLS-DA) was performed to separate metabolic profiles between different groups. The variable importance in the projection (VIP) method in OPLS-DA was used to determine the importance of each metabolite. KEGG pathway enrichment analysis with the differential metabolites was

performed using MetaboAnalyst 5.0. T-tests, Kruskal–Wallis tests and ANOVA were used to evaluate the differences among groups. Correlations between metabolites and clinical data were tested by Pearson's or Spearman's correlation coefficient. Principal component analysis (PCA) was performed by the R packages FactoMineR and factoextra. P <0.05 was regarded as statistically significant.

Results:

Serum metabolomic profiles in untreated SSc patients

Untargeted metabolic analysis was performed by HPLC-Q-TOF-MS/MS to profile the serum metabolites of SSc patients. After excluding metabolites with more than 5% missing data, 1007 metabolic features were identified in positive ion mode and 479 metabolic features were identified in negative ion mode. For both positive and negative modes, OPLS-DA demonstrated a clear separation between HC and untreated SSc patients (SSc baseline), as shown in the 3D score plots (Figure 1A-B) and 2D score plots (Figure 1C-D). Univariate analysis identified 375 elevated metabolites and 471 decreased metabolites in the SSc baseline group (Figure 1E). A total of 164 metabolites (FC >1.2 or <0.83, p<0.05 and VIP>1.0) were further selected for KEGG pathway enrichment analysis. In the serum of untreated SSc patients, upregulated metabolites were enriched in amino acid metabolism such as taurine and hypotaurine metabolism, cysteine metabolism and lysine degradation, glycolytic metabolic and related pathways such as Warburg effect, pentose phosphate pathway and pyruvate metabolism, as well as other metabolic pathways such as bile acid biosynthesis. Downregulated metabolites were enriched in amino acid metabolism such as alpha linolenic acid and linoleic acid metabolism, methylhistidine metabolism, methionine metabolism, and betaine metabolism, as well as fatty acid metabolism including fatty acid biosynthesis and beta oxidation of very long chain fatty acids. (Figure 1F). Representative metabolites such as hydroxyisocaproic acid, citric acid, isobutyric acid, phloretin 2'-O-glucuronide, and thromboxane A2 were substantially enhanced in the SSc baseline group (Figure 1G). Conversely, 5 methoxytryptophol, 12(13)Ep-9-KODE, alpha-tocotrienol, chlorogenic acid, and cholic acid glucuronide were decreased in the SSc baseline group (Figure 1H). Together, these results suggest that SSc patients exhibit a different metabolic profile compared to that to HC.

Serum metabolomic profiles in treated SSc patients

The influence of treatment on metabolomic profiles was examined by multivariate analysis and univariate analysis. Treated SSc patients (SSc treatment) were clearly separated from untreated SSc patients (SSc baseline) in the OPLS-DA 3D score plots (Figure 2A-B) and 2D score plots (Figure 2C-D). A total of 225 elevated metabolites and 281 decreased metabolites were identified by univariate analysis (Figure 2E). Given FC >1.2 or <0.83, p<0.05 and VIP>1.0, 105 increased and 107 decreased metabolites were selected as differential metabolites for KEGG

pathway enrichment analysis. These upregulated metabolites were enriched in the glycometabolism including lactose degradation gluconeogenesis and glucose-alanine cycle, amino acid metabolism including phenylalanine and tyrosine metabolism, glycine and serine metabolism and betaine metabolism, as well as other metabolic pathways such as urea cycle and pyruvaldehyde degradation. Downregulated metabolites were enriched in organic acid metabolism such as alpha linolenic acid and linoleic acid metabolism and citric acid cycle, amino acid metabolism such as methylhistidine metabolism and homocysteine degradation, in addition to other metabolic pathways such as phosphatidylcholine biosynthesis, biotin metabolism, and ubiquinone Biosynthesis (Figure 2F). Serum levels of betaine, gamma-glutamylleucine, glucosylgalactosyl hydroxylysine, L-methionine, and PC(14:1(9Z)/16:1(9Z)) were increased upon treatment (Figure 2G). However, 2-oxoarginine, 3'-hydroxy-e,e-caroten-3-one, 5'-deoxyadenosine, FAPy-adenine, and nicotine were decreased upon treatment (Figure 2H). To investigate the metabolomic reprogramming that occurs during the therapy process, vertical comparisons were conducted with the serum metabolites in untreated SSc patients and treated SSc patients followed up at different time points (1, 3, 6, 12, and 24 months after starting treatment). Metabolites such as amidosulfonic acid, L-proline, L-glutamic acid and betaine, gradually increased close to normal levels upon treatment (Supplementary Figure 1A). Metabolites such as uracil, phthalic acid, guanidinosuccinic acid, and isovalerylglycine, progressively declined to normal levels along with treatment (Supplementary Figure 1B). Collectively, these results show that treatment dynamically affects the metabolic profile of SSc patients.

Partial recovery of serum metabolites in SSc patients after treatment

Our previous analysis confirmed that serum metabolites and pathways were altered in SSc patients. To further identify which metabolites and pathways returned to normal after treatment, we compared the metabolic profiles in the HC, untreated SSc patients (SSc baseline) and treated SSc patients (SSc treatment). These metabolites and pathways may be closely related to the etiology and pathogenesis of SSc. OPLS-DA clearly separated HC, SSc baseline and SSc treatment groups (Figure 3A-B). Functional analysis identified pathways including bile acid biosynthesis, starch and sucrose metabolism, cysteine metabolism, ethanol degradation, citric acid cycle, pyruvate metabolism and Warburg effect that were significantly enhanced in new-onset SSc patients but declined after treatment (Figure 3C). Pathways such as arginine and proline metabolism, androgen and estrogen metabolism, tryptophan metabolism, methionine metabolism, spermidine and spermine biosynthesis, betaine metabolism, and fatty acid biosynthesis, decreased in new-onset SSc patients, but increased after treatment (Figure 3D). Figure 3E shows a panel of metabolites that were elevated in untreated SSc patients, but decreased after treatment. These metabolites included phloretin 2'-O-glucuronide, retinoyl b-glucuronide, and isovalerylglycine (Figure 3G). Representative metabolites that were decreased in SSc patients, but elevated after treatment included all-trans-retinoic acid, betaine, and L-glutamic acid (Figure 3F, 3H). Meanwhile, according to the changes of mRSS by 20% from baseline, the 57 follow-up SSc patients were classified as skin improved (45.61%), skin

progressive (29.82%) and skin stable group (24.56%)(Supplementary Figure 2A). The Changes of metabolites in the three groups were also analyzed (Supplementary Figure 2B). Metabolites including gamma-CEHC, paraxanthine, PS(18:0/18:1(9Z)) were decreased in SSc baseline, but elevated in the skin improved SSc group. 2,3-Diaminosalicylic acid, MG(0:0/18:2(9Z,12Z)/0:0), and phloretin 2'-O-glucuronide were elevated in untreated SSc patients, but decreased in the skin improved SSc group. However, these metabolites have no significant changes in the skin progressive group compared with the SSc baseline level (Supplementary Figure 2C-D). Together, these results identified a set of metabolites and pathways were normalized in response to treatment in SSc, indicating that these metabolites may thus serve as markers of treatment exposure.

Serum metabolomic profiles in fibrotic mouse models

To investigate the metabolomic profiles of fibrotic mouse models, untargeted metabolic analysis was performed with the serum of 11 control mice, 11 mice with BLM-induced fibrosis and 11 mice with HOCl-induced fibrosis. OPLS-DA revealed that mice with BLM-induced fibrosis and mice with HOCl-induced fibrosis were clearly separated from control mice (Figure 4A-B, Supplementary Figure 3A-D, Supplementary Figure 4A-D). Univariate analysis of the metabolites in the two mouse models was performed (Supplementary Figure 3E, Supplementary Figure 4E). There was a strong correlation between the changes in metabolites in these two mouse models, as shown in Figure 4C (showing all detected metabolites) and Figure 4D (showing only metabolites with FC >1.2 or <0.83, p<0.1). Given FC >1.2 or <0.83 and *p*<0.1, 97 metabolites were elevated, and 68 metabolites were decreased concordantly in both mouse models (Figure 4E). KEGG pathway enrichment analysis was performed with these altered metabolites (VIP >1). As shown in Figure 4F, enhanced pathways included thiamine metabolism, lysine degradation, malate-aspartate shuttle, pyruvaldehyde degradation, and transfer of acetyl groups into mitochondria; decreased pathways included catecholamine biosynthesis, betaine metabolism, D-arginine and D-ornithine metabolism, oxidation of branched chain fatty acids, and phytanic acid peroxisomal oxidation. Enrichment analyses were also conducted with differential metabolites in mice with BLM-induced fibrosis and mice with HOCl-induced fibrosis (Supplementary Figure 3F, Supplementary Figure 4F). Most of the pathways changed simultaneously in both mouse models. Serum levels of 1-methyluric acid, LysoPC(16:1(9Z)/0:0), LysoPC(22:1(13Z)/0:0), and sulfate were increased in both BLM-induced and HOCl-induced fibrosis (Figure 4G). 4-Vinylohenol sulfate, allysine, L-tryptophan and uracil were decreased in both the BLM-induced and HOCl-induced fibrosis (Figure 4H). Together, these results suggested that metabolomic profiles were reprogrammed in mice with BLM-induced fibrosis and HOCl-induced fibrosis. In addition, the variations in metabolites and pathways were relatively consistent in these two mouse models.

Altered pathways and metabolites identified in both SSc patients and fibrotic mouse models

To identify key pathways and metabolites, an integrated analysis of serum metabolomic profiles in both SSc patients and fibrotic mouse models was performed. Common pathways

enhanced in both patients and mouse models included organic acid metabolism such as arachidonic acid metabolism and citric acid cycle, glycolytic metabolic and related pathways such as pentose phosphate pathway and warburg effect, amino acid metabolisms such as cysteine metabolism, lysine degradation and taurine and hypotaurine metabolism, as well as other metabolic pathways such as transfer of acetyl groups into mitochondria and bile acid biosynthesis (Figure 5A). Pathways decreased in both SSc patients and mouse models included amino acid metabolism such as arginine and proline metabolism, methionine metabolism, tryptophan metabolism and betaine metabolism, as well as other metabolisms such as fatty acid biosynthesis, nicotinate and nicotinamide metabolism, and purine metabolism (Figure 5B). Eleven serum metabolites including thromboxane B2, PS(18:0/22:5(7Z, 10Z, 13Z, 16Z, 19Z)), isovalerylglycine, LysoPC(16:1(9Z)/0:0) and retinoyl b-glucuronide were upregulated in both SSc patients and mouse models(Figure 5C). Allysine, L-tryptophan, urea and 5-acetylamino-6-formylamino-3-methyluracil were downregulated in SSc patients and mouse models (Figure 5D). These data demonstrated that SSc patients and fibrotic mouse models shared dysregulated pathways and metabolites, suggesting that the shared markers may serve as general biomarkers for fibrotic tissue remodeling.

Correlation analysis of the metabolites and clinical parameters

To identify the association of serum metabolites with clinical parameters, correlation analyses were performed for important metabolites that changed in SSc patients. Multiple metabolites demonstrated significant correlations with the clinical parameters of SSc patients (Supplementary Figure 5A). For example, L-Tryptophan, a decreased serum metabolite in SSc, was negatively correlated with inflammatory markers such as the erythrocyte sedimentation rate (ESR), C-reactive protein (CRP), interleukin (IL)-6 and tumor necrosis factor (TNF)-α (Supplementary Figure 5A). All-trans-retinoic acid, which is decreased in SSc patients and returns to normal levels after treatment, was negatively associated with mRSS, ESR, and IgG (Supplementary Figure 5A). Betaine was decreased in SSc, but positively associated with serum IL-6 (Supplementary Figure 5A). The increased metabolite thromboxane B2 was negatively associated with the SSc-specific symptom sclerodactyly, but positively associated with abnormal nailfold capillaries, TNFα, and complement C3 (Supplementary Figure 5A). Phthalic acid was also elevated in SSc and positively associated with sclerodactyly (Supplementary Figure 5A). Special attention was given to the metabolites associated with mRSS and ILD. As shown in Figures 6A and 6B, allysine and all-trans-retinoic acid were negatively associated, while D-glucuronic acid and hexanoyl carnitine were positively associated with the mRSS. PCA recognized a panel of metabolites that can effectively distinguish ILD from non-ILD patients (Figure 6C). Metabolites such as gamma-linolenic acid, dihydrothymine, etiocholanolone glucuronide, L-pipecolic acid, carnosine and L-cystathionine were elevated in SSc patients with ILD. Proline betaine, androsterone sulfate, dihydrothymine, phloretin 2'-O-glucuronide, 4-guanidinobutanoic acid, and NNAL-N-glucuronide were decreased in SSc patients with ILD (Figure 6D, Supplementary Figure 5B). To evaluate the potential of metabolites to predict the progression of skin fibrosis in SSc patients, we constructed 10 ML classifiers. According to the

accuracy score, the random forest was chosen as the optimal performance classifier (Figure 6E). RFE in the random forest model selected the most important features in predicting the skin progression of SSc patients. LDA was performed based on the top 16 important features. As shown in the LDA plot, SSc patients with skin improved, stable and progressive were clearly separated (Figure 6F).These metabolites includes medicagenic acid 3-O-b-D-glucuronide, 4'-O-methyl-(-)-epicatechin-3'-O-beta-glucuronide, valproic acid glucuronide, etc (Supplementary Table 3). These results suggested that there are strong correlations between serum metabolites and clinical parameters of SSc patients. Some metabolites could be used as biomarkers to predict SSc ILD and the progression of skin fibrosis.

Discussion

Metabolic perturbations participate in the initiation and progression of SSc. In our study, the largest metabolic analysis in SSc with unbiased serum profiling was performed. A total of 127 baselines newly diagnosed SSc samples and 57 treated SSc samples (follow-up samples of the 127 newly diagnosed SSc patients) were included in our study. We comprehensively analyzed the metabolomic profiles of SSc patients and fibrotic mouse models, as well as the metabolic alterations in SSc patients before and after treatment. Diverse metabolites and metabolic pathways associated with inflammation and fibrosis have also been identified.

Some of the amino acids and relative metabolic pathways involved in the inflammatory response in SSc were dysregulated, including L-tryptophan metabolism, taurine and hypotaurine metabolism methionine metabolism, and betaine metabolism. Tryptophan is an essential amino acid and is metabolized mainly through the kynurenine pathway (KP)[30, 31]. Tryptophan metabolism and the KP inhibit inflammation by suppressing T-cell and NK-cell proliferation[32]. A reduction in L-tryptophan was found in SSc patients and in both BLM- and HOCl-induced fibrotic mouse models, and the level of L-tryptophan in SSc patients was negatively correlated with inflammatory markers such as IL-6, consistent with the findings in COVID-19 patients[33]. Taurine and hypotaurine metabolism were elevated in both SSc patients and mouse models. Previous studies confirmed that taurine and hypotaurine metabolism was involved in oxidative stress and inflammation[34]. Patients with gout and hyperuricemia also exhibited altered taurine and hypotaurine metabolism[35]. Methionine metabolism was found to be reduced in both SSc patients and mouse models. Moreover, L-methionine was restored in SSc patients after treatment. Methionine oxidation and reduction commonly occurred under oxidative stress conditions[36]. Activated $CD4^+$ T cells consumed methionine to synthesize S-adenosyl-L-methionine (SAM), which is the main methyl donor[37]. Th17 cells also need to take up methionine to maintain cell proliferation and cytokine production[38]. Methionine-deficient diet aggravated hepatic steatosis and significantly increased the production of proinflammatory cytokines such as IL-6, IL-1β and TNF-α[39]. Betaine, also known as trimethylglycine, was significantly reduced in patients with new-onset SSc, but returned to normal levels after treatment. Multiple studies have confirmed that betaine can exert anti-inflammatory,

antioxidative stress and antiapoptotic effects by inhibiting NF-κB activity, NLRP3 inflammasome activation and Akt/mTOR signaling[40].

Dysregulated glycolytic metabolic and related pathways were also strongly associated with the breakdown of immune homeostasis in the pathogenesis of SSc. The Warburg effect, pyruvate acid metabolism, and the pentose phosphate pathway were upregulated in SSc patients and fibrosis mouse models. The Warburg effect, or aerobic glycolysis, was first identified in rapidly growing and proliferating tumor cells. It was later found that the shift from oxidative phosphorylation to aerobic glycolysis is a hallmark of pro-inflammatory immune responses, such as activation of effector T cells and polarization of M1 macrophages[41]. The pentose phosphate pathway (PPP), which generates essential intermediates for nucleotide and amino acid synthesis, is essential for the proliferation and survival of immune cells[42]. However, a recent study found that the PPP is involved in the prevention of autoimmunity by promoting the function of regulatory T cells[43]. And increase in the PPP in SSc may be secondary to the enhanced anabolism or activation of autoimmune responses.

Triggered by inflammation and autoimmunity, skin and visceral fibrosis are prominent in patients with advanced SSc. Moreover, fibrosis is a central feature and a major cause of organ dysfunction in SSc patients. The metabolic signatures of fibrotic tissue remodeling shared by humans and mice were also identified in our study, suggesting a general metabolic alteration underlying the fibrosis mechanism.

Multiple dysregulated organic acids found in this study were associated with fibrosis. All-trans retinoic acid (ATRA), an active metabolite of vitamin A, was downregulated in new-onset SSc patients, but recovered upon treatment. In our study, serum levels of ATRA were negatively correlated with mRSS in patients with SSc. ATRA can be used in the treatment of acne, psoriasis, ichthyosis and other skin diseases because it modulates the expression of keratin[44]. Previous studies have identified that ATRA can alleviate visceral fibrosis in a TGF-β1/Smad3 signaling dependent or independent manner[45]. ATRA-treated unilateral ureteral obstruction (UUO) rats exhibited attenuated renal interstitial fibrosis and reduced TGF-β1, FN, and Col-IV expression[46]. ATRA administration attenuated irradiated or BLM-induced pulmonary fibrosis by suppressing NF-κB, TGF-β1/Smad3, EphA2-EphrinA1 or PI3K-Akt signaling[47-50]. Moreover, retinoic acid inhibited skin fibrosis in patients with SSc and morphea[51, 52]. L-Pipecolic acid was significantly elevated in SSc patients with ILD. Furthermore, it was also increased in the plasma of patients with liver cirrhosis[53]. However, the specific role and mechanism of L-pipecolic acid in fibrosis are still unclear. Citric acid and the citric acid cycle were elevated in SSc patients and decreased after treatment. Citric acid is an intermediate in the tricarboxylic acid (TCA) cycle. TGF-β promotes the entry of glucose and the activity of the TCA cycle[54]. Phthalic acid increased in SSc patients and was positively associated with skin fibrosis. The increase in phthalic acid in biofluids or tissues is generally due to exposure to phthalate products, which may be an environmental risk factor for SSc. In addition to the metabolites mentioned above, there are

several other organic acids that also change in SSc, including guanidinosuccinic acid, and hippuric acid. However, further studies on the relationship between these organic acids and fibrosis are needed.

In addition, our study identified altered lipid metabolism, which may also contribute to fibrosis in SSc. Sphingosine 1-phosphate (S1P), a metabolic product of sphingolipids, was found to be increased in the serum of SSc patients (data not shown). S1P binds to S1P receptors (S1PRs) to exert its function. Disturbed S1PR signaling on fibroblasts, macrophages, and endothelial cells contributes to tissue fibrosis, inflammation and vascular development in distinctive diseases[55]. Thus, S1P or its receptors could be a promising therapeutic target for SSc. Lysophospholipids were also dysregulated in SSc. We identified increased LysoPC(18:1(11Z)) and decreased LysoPC(22:2(13Z,16Z)) in new-onset SSc patients, and their levels returned to normal upon treatment. Previous studies have found that lysophospholipids and lysophosphatidic acid (LPA) are related to fibrosis and inflammation[56, 57]. Moreover, elevated LPA was also identified in SSc, and it may participate in the pathogenesis of SSc by binding to peroxisome proliferation activated receptor γ (PPARγ)[58]. LPA1 antagonists have entered early phase clinical trials in SSc and idiopathic pulmonary fibrosis. Distinctive phosphatidylcholines, such as PC(14:1(9Z)/16:1(9Z)) and PC(16:0/P-18:1(9Z)), were identified to be downregulated in patients with new-onset SSc and were elevated to normal levels upon treatment. Previous studies have demonstrated that phosphatidylcholine protects against organ fibrosis in cirrhosis and cystic fibrosis[59, 60].

In summary, our study accomplished the largest metabolic analysis in SSc with unbiased serum profiling. A wide range of dysregulated metabolites and related pathways have been identified. Moreover, a set of metabolites was shown to be strongly associated with clinical features such as ILD and skin fibrosis. Based on the performance of the ML algorithm, the most important metabolites that can predict the progression of mRSS are identified. Our study provides novel insights into the mechanisms underlying the development of SSc, as well as a set of metabolic biomarker candidates for SSc. However, there are some limitations to this study. Firstly, it was a single-center study conducted among Chinese individuals. Therefore, validation in independent cohorts with different genetic backgrounds is necessary. Secondly, an expanded cohort of follow-up patients is required to confirm the reprogramming of metabolomics during treatment. Lastly, additional investigations are necessary to explore the mechanisms underlying the metabolic changes observed in SSc.

Conclusion

In summary, our research shows that both SSc patients and fibrotic mouse models have a disturbed metabolic profile. Differentially expressed metabolites and dysregulated metabolic pathways were recognized. The metabolic alterations in SSc patients and mouse models overlapped to some extent, indicating a general metabolic change associated with fibrotic tissue remodeling in SSc. By comparing new-onset and follow-up patients, patterns of changes

in metabolomics during treatment were identified in this study. Furthermore, we found that treatment partially corrected the metabolic changes in SSc. Several metabolic changes were associated with the clinical parameters in patients with SSc. In addition, certain metabolites were verified to be associated with skin fibrosis and ILD. ML algorithms clearly separated SSc patients with progressive mRSS and identified important metabolites that can predict the exacerbation of skin fibrosis. Metabolic changes in SSc serum may thus serve as novel biomarker candidates in SSc.

Abbreviations

SSc	Systemic Sclerosis
mRSS	modified Rodnan skin score
ILD	Interstitial Lung Disease
ML	Machine learning
HPLC-Q-TOF-MS	High-Performance Liquid Chromatography of Quadrupole Time-of-Flight Mass Spectrometry
LC–MS	Liquid Chromatography–Mass Spectrometry
BLM	Bleomycin
HC	Healthy Controls
OPLS-DA	Orthogonal Partial Least-Squares Discriminate Analysis
QC	Quality Controls
FC	Fold Change
VIP	Variable Importance in the Projection
PCA	Principal Component Analysis
ESR	Erythrocyte Sedimentation Rate
CRP	C-Reactive Protein
IL	Interleukin
TNF	Tumor Necrosis Factor
UA	Uric Acid
PT	Prothrombin Time
INR	International Normalized Ratio
APTT	Activated Partial Thromboplastin Time
KP	kynurenine Pathway

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Acknowledgements

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Conflict of interest

The authors have declared no conflicts of interest.

CRediT authorship contribution statement

Muyao Guo: Conceptualization, Investigation, Methodology, Investigation, Formal analysis, Visualization, Writing - original draft. **Di Liu**: Methodology, Software, Visualization. **Yu Jiang**: Methodology, Software. **Weilin Chen**: Methodology. **Lijuan Zhao**: Methodology, Software. **Ding Bao**: Methodology. **Yisha Li**: Writing - review & editing. **Jörg H.W. Distler**: Conceptualization, Methodology, Writing - review & editing. **Honglin Zhu**: Conceptualization, Supervision, Validation, Data curation, Writing - review & editing.

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Figure 1. Serum metabolomic profiles in untreated SSc patients. **(A-B)** OPLS-DA score scatter 3D plot of serum metabolomics data compared HC and untreated SSc patients (SSc baseline) in positive ion mode (A) and negative ion mode (B). **(C-D)** OPLS-DA score scatter 2D plot of serum metabolomics data compared HC and untreated SSc patients (SSc baseline) in positive ion mode (C) and negative ion mode (D). **(E)** Volcano plot of up-regulated and down-regulated metabolites in untreated SSc patients (SSc baseline). **(F)** KEGG pathway analysis of up-regulated and down-regulated metabolites in untreated SSc patients (SSc baseline). **(G)** Up-regulated metabolites in untreated SSc patients (SSc baseline). **(H)** Down-regulated metabolites in untreated SSc patients (SSc baseline). (*, < 0.05; **, < 0.01; ***, < 0.001, ****, <0.0001).

Figure 2. Serum metabolomic profiles in treated SSc patients. (A-B) OPLS-DA score scatter 3D plot of serum metabolomics data compared untreated SSc patients (SSc baseline) and treated SSc patients (SSc treatment) in positive ion mode (A) and negative ion mode (B). **(C-D)** OPLS-DA score scatter 2D plot of serum metabolomics data compared untreated SSc patients (SSc baseline) and treated SSc patients (SSc treatment) in positive ion mode (C) and negative ion mode (D). **(E)** Volcano plot of up-regulated and down-regulated metabolites in treated SSc patients (SSc treatment) compared with untreated SSc patients (SSc baseline). **(F)** KEGG pathway analysis of up-regulated and down-regulated metabolites in treated SSc patients (SSc treatment). **(G)** Up-regulated metabolites in treated SSc patients (SSc treatment). **(H)** Down-regulated metabolites in treated SSc patients (SSc treatment). (*, < 0.05; **, < 0.01; ***, < 0.001, ****, <0.0001).

Figure 3. Partial recovery of serum metabolites in SSc patients after treatment. (A-B) OPLS-DA score scatter 3D plot of serum metabolomics data compared HC, untreated SSc patients (SSc baseline) and treated SSc patients (SSc treatment) in positive ion mode (A) and negative ion mode (B). **(C)** Pathways that up-regulated before treatment but down-regulated after treatment. **(D)** Pathways that down-regulated before treatment but up-regulated after treatment. **(E)** Heatmap of metabolites that up-regulated before treatment but down-regulated after treatment. **(F)** Heatmap of metabolites that down-regulated before treatment but up-regulated after treatment. **(G)** Metabolites that were up-regulated before treatment and down-regulated after treatment. **(H)** Metabolites that were down-regulated before treatment and up-regulated after treatment. (*, < 0.05; **, < 0.01; ***, < 0.001, ****, <0.0001).

Figure 4. Serum metabolomic profiles in fibrotic mouse models. (A-B) OPLS-DA score scatter 3D plot of serum metabolomics data compared control mice (NaCl), mice with bleomycin (BLM)-induced fibrosis and mice with hypochlorous acid (HOCl)-induced fibrosis in positive ion mode (A) and negative ion mode (B). **(C-D)** Correlation analysis of the metabolomic profiles between BLM-induced fibrosis and HOCl-induced fibrosis. (C) Showing all detected metabolites. (D) Showing only metabolites with FC >1.2 or <0.83, p<0.1. **(E)** Venn diagram of up- and down-regulated metabolites in BLM-induced fibrosis and HOCl-induced fibrosis. **(F)** KEGG pathway analysis of up- regulated and down-regulated metabolites in mice with both BLM-induced fibrosis and HOCl-induced fibrosis. (**G**) Metabolites that were up-regulated in mice with both

BLM-induced and HOCl-induced fibrosis. (**H**) Metabolites that were down-regulated in mice with both BLM-induced and HOCl-induced fibrosis. (*, < 0.05; **, < 0.01; ***, < 0.001, ****, <0.0001).

Figure 5. Altered pathways and metabolites identified in both SSc patients and fibrotic mouse models. **(A)** Up-regulated pathways in both SSc patients and mouse models. **(B)** Down-regulated pathways in both SSc patients and mouse models. **(C)** Up-regulated metabolites in both SSc patients and mouse models. **(D)** Down-regulated metabolites in both SSc patients and mouse models. (VIP **>** 1, FC >1.2 or **<** 0.83).

Figure 6. Correlation analysis of the metabolites and clinical parameters. (A) Metabolites that were negatively associated with mRSS. **(B)** Metabolites that were positively associated with mRSS. **(C)** PCA plot of serum metabolomics compared SSc patients with or without ILD. **(D)** Heatmap of differential metabolites between ILD patients and non-ILD patients. **(E)** 10 ML algorithms used to distinguish patients with mRSS progression (mean and 95% confidential interval). **(F)** The LDA plot showing patients with improved, progressive and stable level of mRSS.

Graphical abstract

Highlights:

1. Systemic sclerosis (SSc) patients and fibrotic mouse models exhibited unique serum metabolic profiles, and shared altered metabolites and metabolic pathways.

2. Treatment partially restored the metabolic changes in SSc, and some metabolites were associated with the treatment response of SSc.

3. Some altered metabolites in SSc patients were associated with clinical parameters such as the modified Rodnan skin score (mRSS) and interstitial lung disease (ILD).

4. Machine learning (ML) algorithms identified important metabolites that may predict the progression of mRSS.

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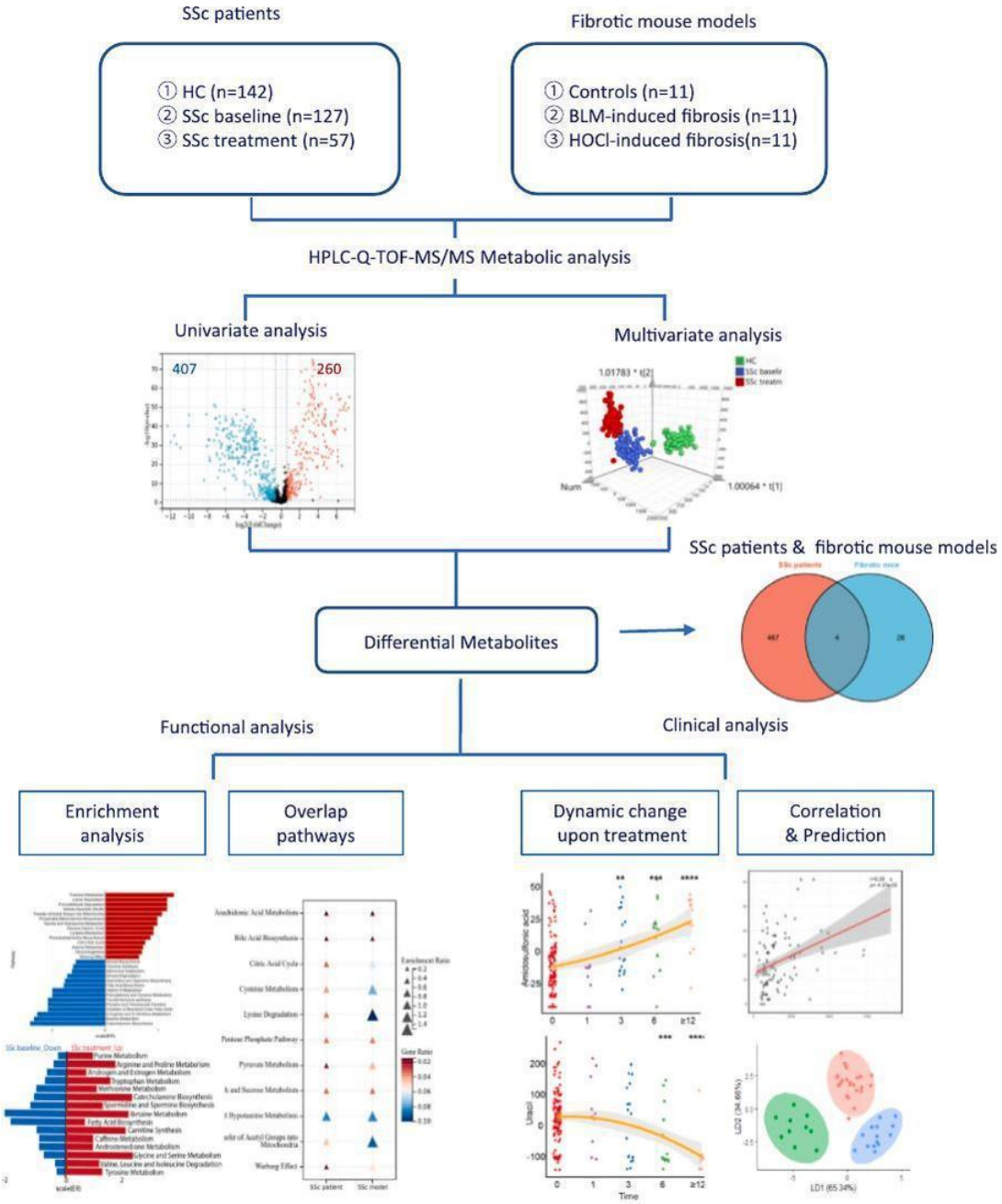
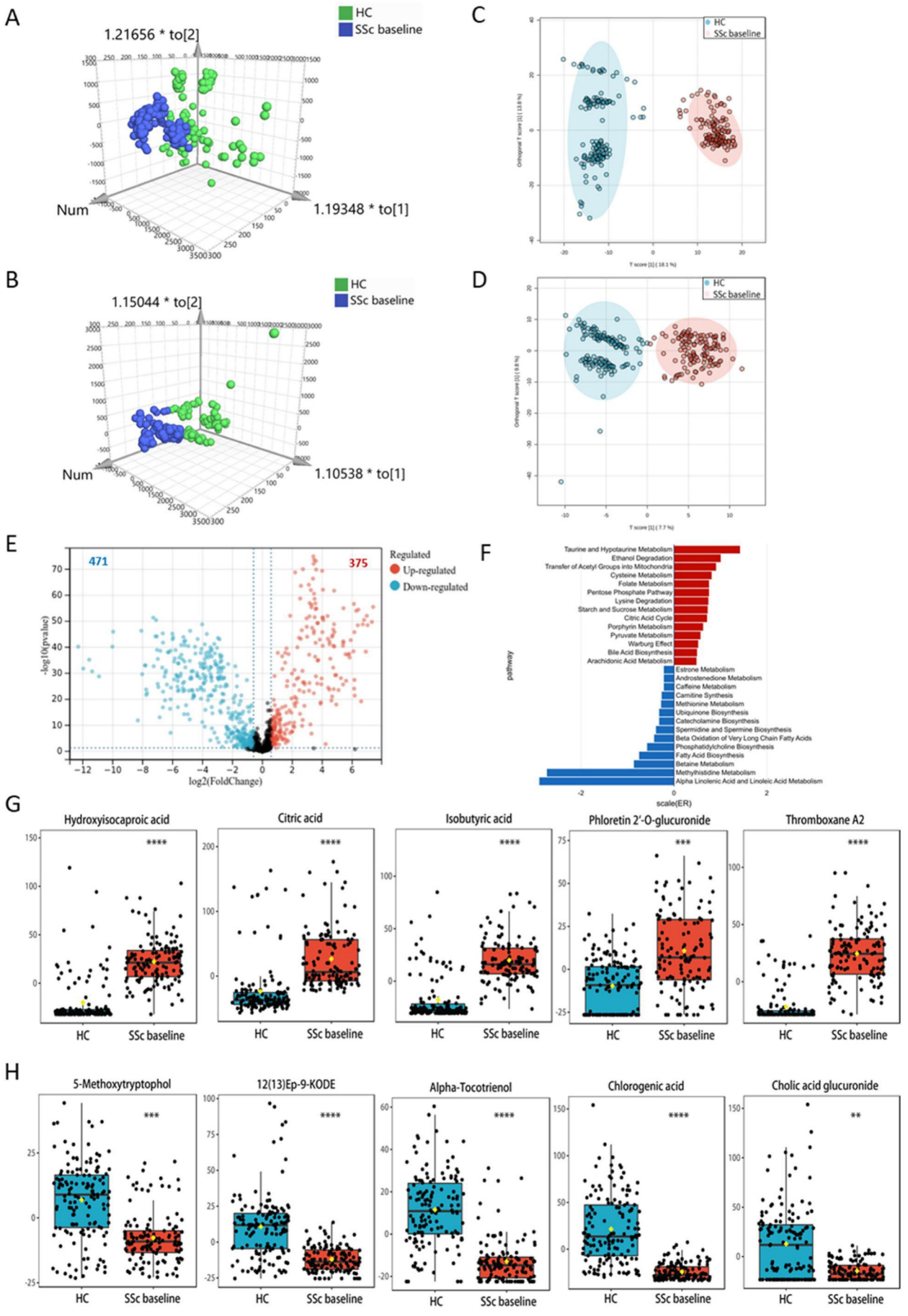


Graphics Abstract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D
HC
SSc baseline
E
471
375
Regulated
Up-regulated
Down-regulated
-log10(pvalue)
log2(FoldChange)
F
Taurine and Hypotaurine Metabolism
Ethanol Degradation
Transfer of Acetyl Groups into Mitochondria
Cysteine Metabolism
Folate Metabolism
Pentose Phosphate Pathway
Lysine Degradation
Starch and Sucrose Metabolism
Citric Acid Cycle
Porphyrin Metabolism
Pyruvate Metabolism
Warburg Effect
Bile Acid Biosynthesis
Arachidonic Acid Metabolism
Estrone Metabolism
Androstenedione Metabolism
Caffeine Metabolism
Carnitine Synthesis
Methionine Metabolism
Ubiquinone Biosynthesis
Catecholamine Biosynthesis
Spermidine and Spermine Biosynthesis
Beta Oxidation of Very Long Chain Fatty Acids
Phosphatidylcholine Biosynthesis
Fatty Acid Biosynthesis
Betaine Metabolism
Methylhistidine Metabolism
Alpha Linolenic Acid and Linoleic Acid Metabolism
pathway
scale(ER)
G
Hydroxyisocaproic acid
Citric acid
Isobutyric acid
Phloretin 2'-O-glucuronide
Thromboxane A2
HC
SSc baseline
H
5-Methoxytryptophol
12(13)Ep-9-KODE
Alpha-Tocotrienol
Chlorogenic acid
Cholic acid glucuronide
HC
SSc baseline

Figure 1

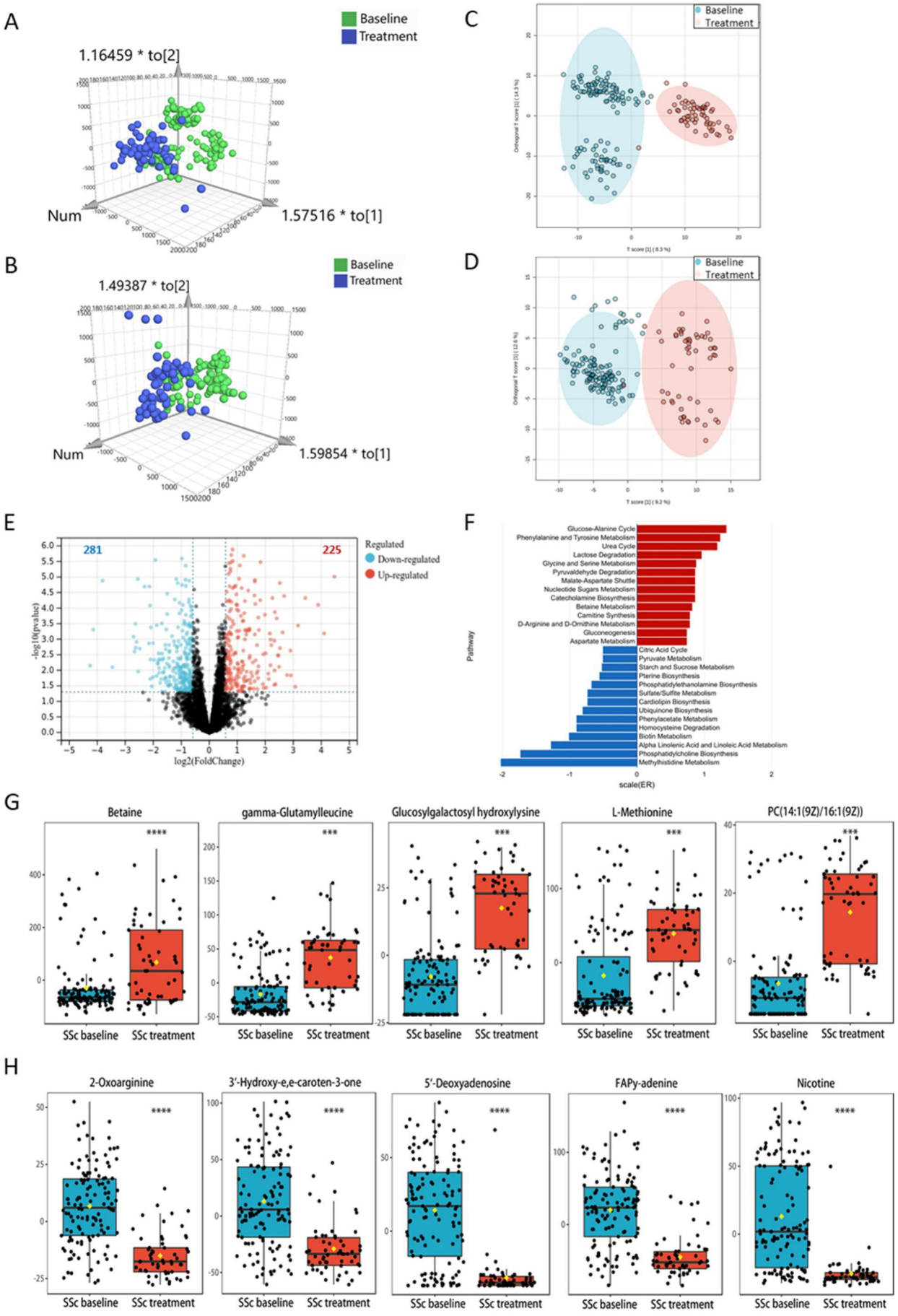

A
Baseline
Treatment
1.16459 * to[2]
Num
1.57516 * to[1]
B
Baseline
Treatment
1.49387 * to[2]
Num
1.59854 * to[1]
C
Baseline
Treatment
D
Baseline
Treatment
E
281
225
Regulated
Down-regulated
Up-regulated
-log10(pvalue)
log2(FoldChange)
F
Glucose-Alanine Cycle
Phenylalanine and Tyrosine Metabolism
Urea Cycle
Lactose Degradation
Glycine and Serine Metabolism
Pyruvaldehyde Degradation
Malate-Aspartate Shuttle
Nucleotide Sugars Metabolism
Catecholamine Biosynthesis
Betaine Metabolism
Carnitine Synthesis
D-Arginine and D-Ornithine Metabolism
Gluconeogenesis
Aspartate Metabolism
Citric Acid Cycle
Pyruvate Metabolism
Starch and Sucrose Metabolism
Pterine Biosynthesis
Phosphatidylethanolamine Biosynthesis
Sulfate/Sulfite Metabolism
Cardiolipin Biosynthesis
Ubiquinone Biosynthesis
Phenylacetate Metabolism
Homocysteine Degradation
Biotin Metabolism
Alpha Linolenic Acid and Linoleic Acid Metabolism
Phosphatidylcholine Biosynthesis
Methylhistidine Metabolism
Pathway
scale(ER)
G
Betaine
gamma-Glutamylleucine
Glucosylgalactosyl hydroxylysine
L-Methionine
PC(14:1(9Z)/16:1(9Z))
SSc baseline
SSc treatment
H
2-Oxoarginine
3'-Hydroxy-e,e-caroten-3-one
5'-Deoxyadenosine
FAPy-adenine
Nicotine
SSc baseline
SSc treatment

Figure 2

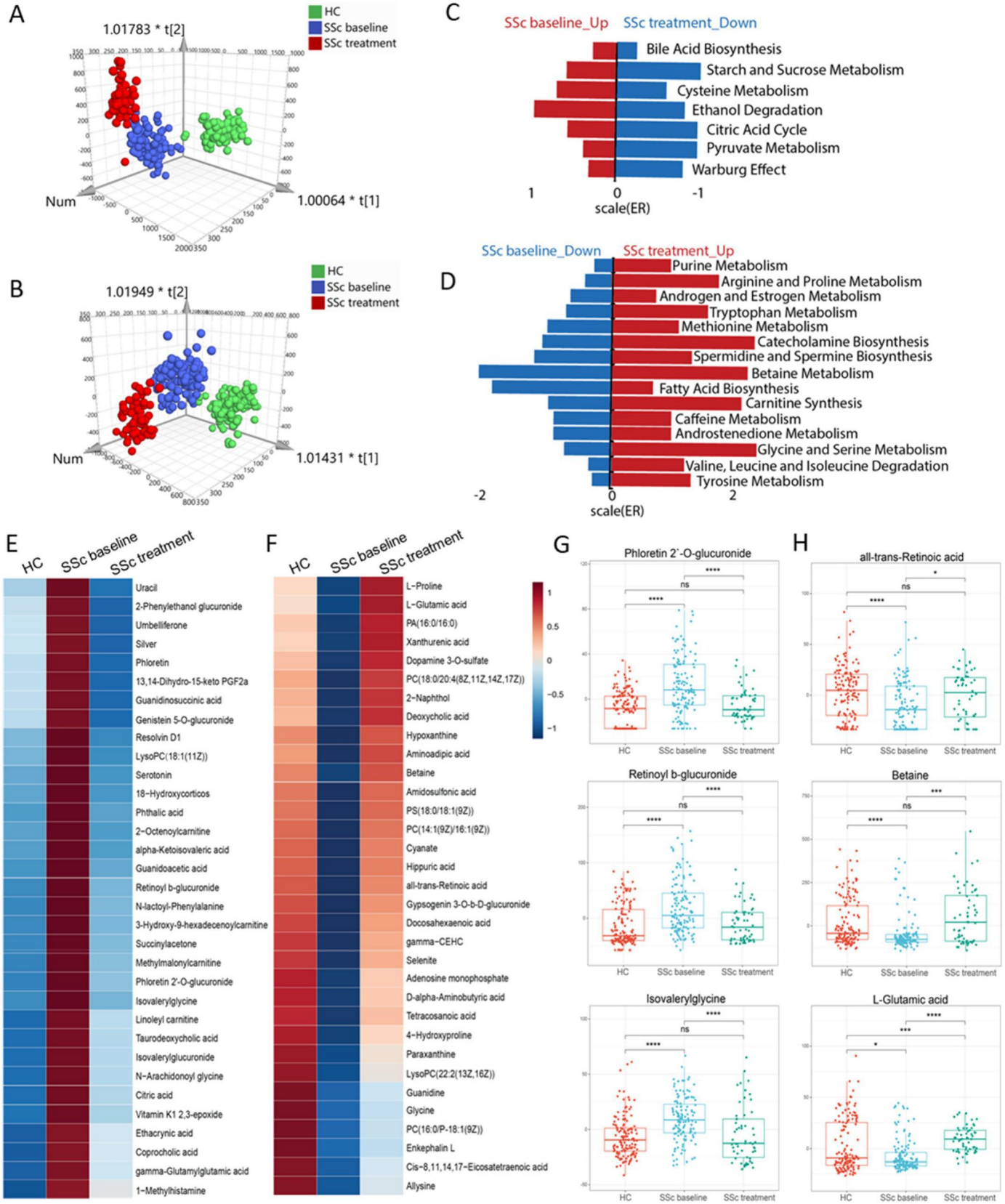

A
1.01783 * t[2]
HC
SSc baseline
SSc treatment
Num
1.00064 * t[1]
B
1.01949 * t[2]
HC
SSc baseline
SSc treatment
Num
1.01431 * t[1]
C
SSc baseline_Up
SSc treatment_Down
Bile Acid Biosynthesis
Starch and Sucrose Metabolism
Cysteine Metabolism
Ethanol Degradation
Citric Acid Cycle
Pyruvate Metabolism
Warburg Effect
1
0
-1
scale(ER)
D
SSc baseline_Down
SSc treatment_Up
Purine Metabolism
Arginine and Proline Metabolism
Androgen and Estrogen Metabolism
Tryptophan Metabolism
Methionine Metabolism
Catecholamine Biosynthesis
Spermidine and Spermine Biosynthesis
Betaine Metabolism
Fatty Acid Biosynthesis
Carnitine Synthesis
Caffeine Metabolism
Androstenedione Metabolism
Glycine and Serine Metabolism
Valine, Leucine and Isoleucine Degradation
Tyrosine Metabolism
-2
0
2
scale(ER)
E
HC
SSc baseline
SSc treatment
Uracil
2-Phenylethanol glucuronide
Umbelliferone
Silver
Phloretin
13,14-Dihydro-15-keto PGF2a
Guanidinosuccinic acid
Genistein 5-O-glucuronide
Resolvin D1
LysoPC(18:1(11Z))
Serotonin
18-Hydroxycorticos
Phthalic acid
2-Octenoylcarnitine
alpha-Ketoisovaleric acid
Guanidoacetic acid
Retinoyl b-glucuronide
N-lactoyl-Phenylalanine
3-Hydroxy-9-hexadecenoylcarnitine
Succinylacetone
Methylmalonylcarnitine
Phloretin 2'-O-glucuronide
Isovalerylglycine
Linoleyl carnitine
Taurodeoxycholic acid
Isovalerylglucuronide
N-Arachidonoyl glycine
Citric acid
Vitamin K1 2,3-epoxide
Ethacrynic acid
Coprocholic acid
gamma-Glutamylglutamic acid
1-Methylhistamine
F
HC
SSc baseline
SSc treatment
L-Proline
L-Glutamic acid
PA(16:0/16:0)
Xanthurenic acid
Dopamine 3-O-sulfate
PC(18:0/20:4(8Z,11Z,14Z,17Z))
2-Naphthol
Deoxycholic acid
Hypoxanthine
Aminoadipic acid
Betaine
Amidosulfonic acid
PS(18:0/18:1(9Z))
PC(14:1(9Z)/16:1(9Z))
Cyanate
Hippuric acid
all-trans-Retinoic acid
Gypsogenin 3-O-b-D-glucuronide
Docosahexaenoic acid
gamma-CEHC
Selenite
Adenosine monophosphate
D-alpha-Aminobutyric acid
Tetracosanoic acid
4-Hydroxyproline
Paraxanthine
LysoPC(22:2(13Z,16Z))
Guanidine
Glycine
PC(16:0/P-18:1(9Z))
Enkephalin L
Cis-8,11,14,17-Eicosatetraenoic acid
Allysine
1
0.5
0
-0.5
-1
G
Phloretin 2'-O-glucuronide
Retinoyl b-glucuronide
Isovalerylglycine
H
all-trans-Retinoic acid
Betaine
L-Glutamic acid
HC
SSc baseline
SSc treatment
ns
****
***
*

Figure 3

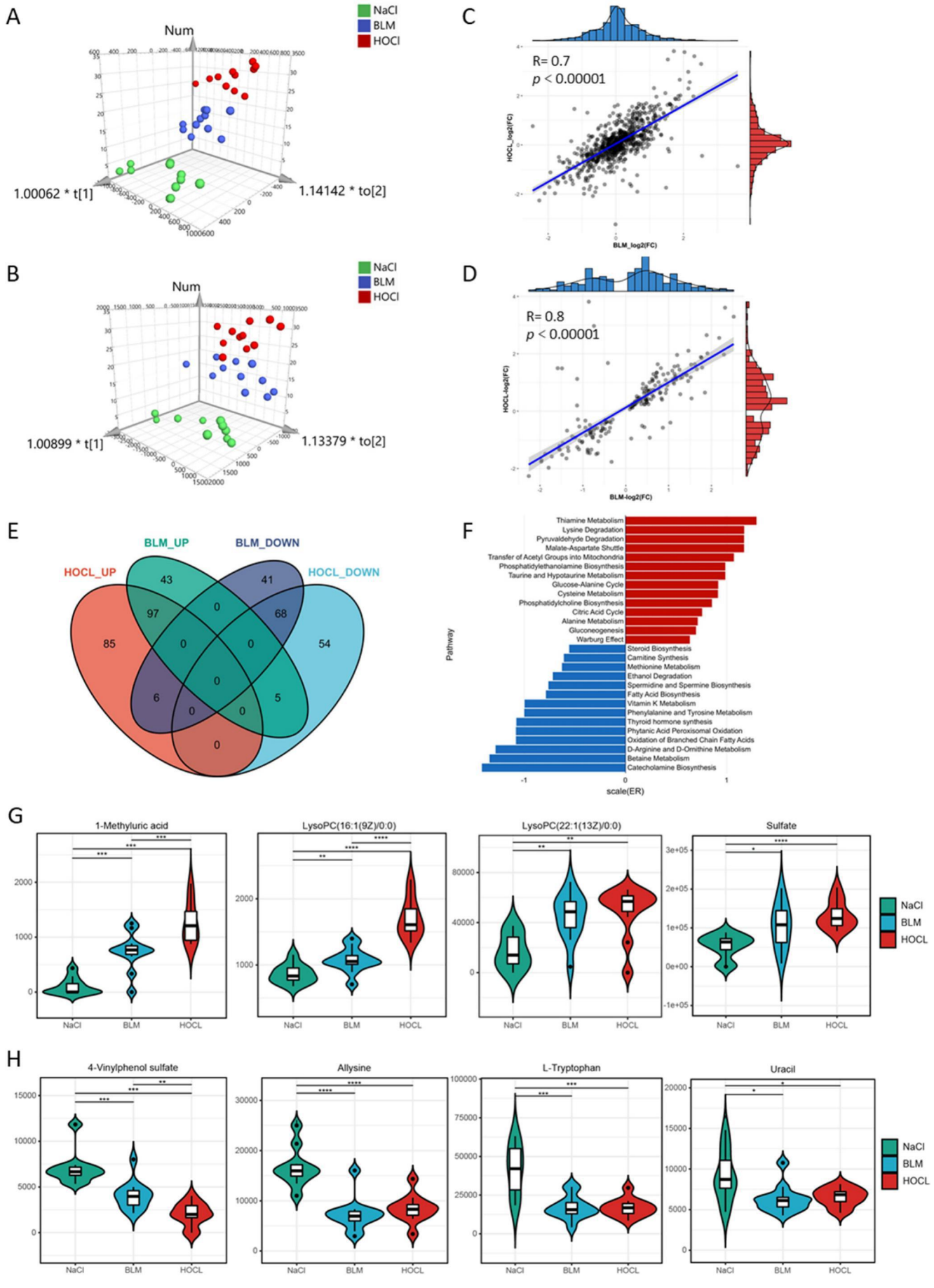
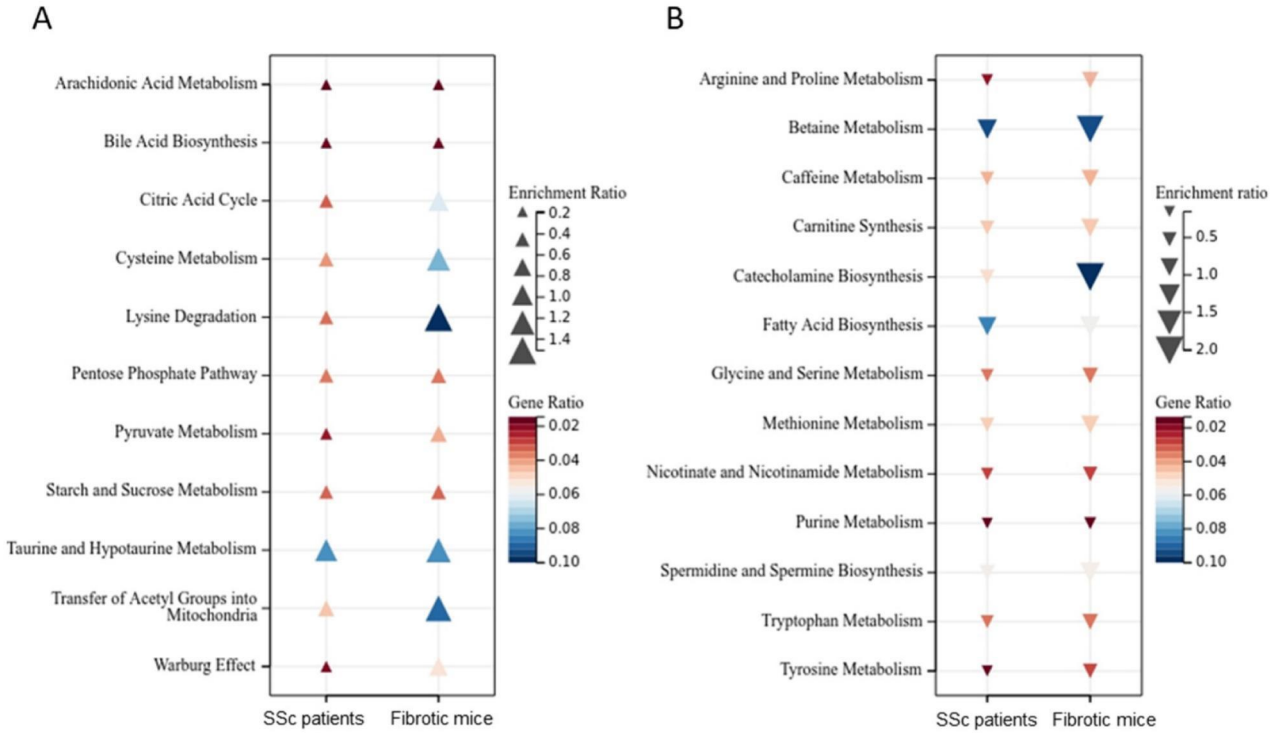
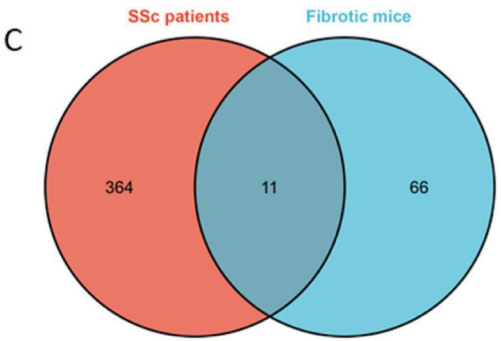
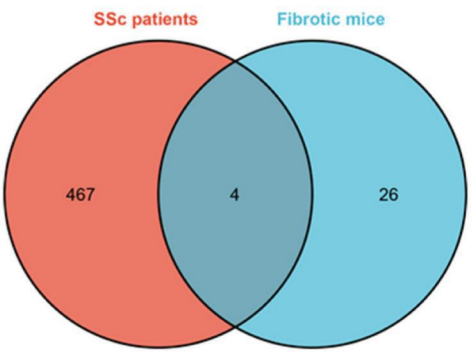

A
NaCl
BLM
HOCl
Num
1.00062 * t[1]
1.14142 * to[2]
B
NaCl
BLM
HOCl
Num
1.00899 * t[1]
1.13379 * to[2]
C
R= 0.7
p < 0.00001
HOCL_log2(FC)
BLM_log2(FC)
D
R= 0.8
p < 0.00001
HOCL-log2(FC)
BLM-log2(FC)
E
BLM_UP
BLM_DOWN
HOCL_UP
HOCL_DOWN
43
41
97
0
68
85
0
0
54
0
6
5
0
0
0
F
Thiamine Metabolism
Lysine Degradation
Pyruvaldehyde Degradation
Malate-Aspartate Shuttle
Transfer of Acetyl Groups into Mitochondria
Phosphatidylethanolamine Biosynthesis
Taurine and Hypotaurine Metabolism
Glucose-Alanine Cycle
Cysteine Metabolism
Phosphatidylcholine Biosynthesis
Citric Acid Cycle
Alanine Metabolism
Gluconeogenesis
Warburg Effect
Steroid Biosynthesis
Carnitine Synthesis
Methionine Metabolism
Ethanol Degradation
Spermidine and Spermine Biosynthesis
Fatty Acid Biosynthesis
Vitamin K Metabolism
Phenylalanine and Tyrosine Metabolism
Thyroid hormone synthesis
Phytanic Acid Peroxisomal Oxidation
Oxidation of Branched Chain Fatty Acids
D-Arginine and D-Ornithine Metabolism
Betaine Metabolism
Catecholamine Biosynthesis
Pathway
scale(ER)
G
1-Methyluric acid
LysoPC(16:1(9Z)/0:0)
LysoPC(22:1(13Z)/0:0)
Sulfate
NaCl
BLM
HOCL
H
4-Vinylphenol sulfate
Allysine
L-Tryptophan
Uracil
NaCl
BLM
HOCL

Figure 4



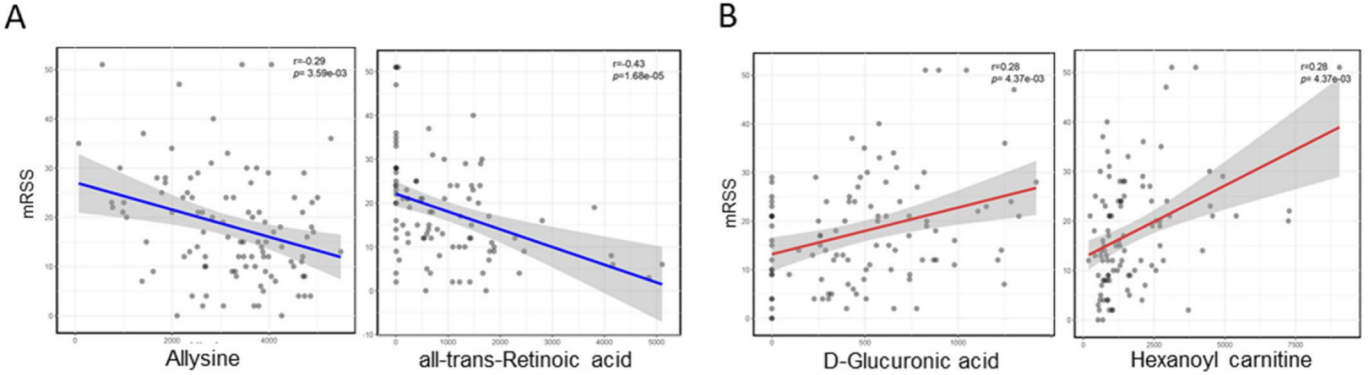


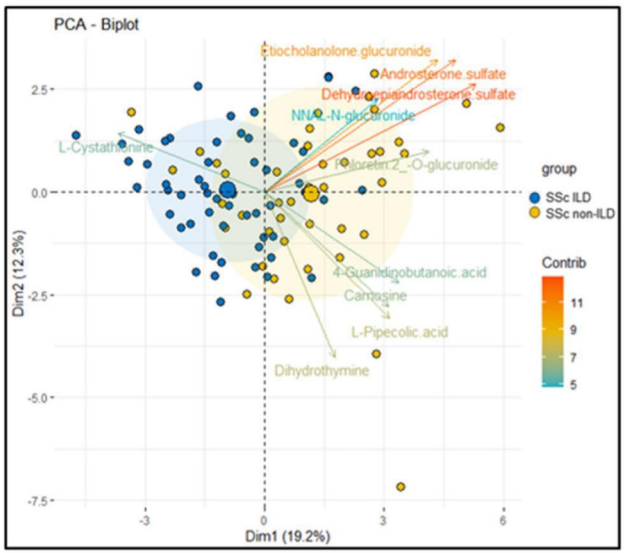
Thromboxane B2
PS(18:0/22:5(7Z,10Z,13Z,16Z,19Z))
Isovalerylglycine
LysoPC(16:1(9Z)/0:0)
Retinoyl b-glucuronide
...

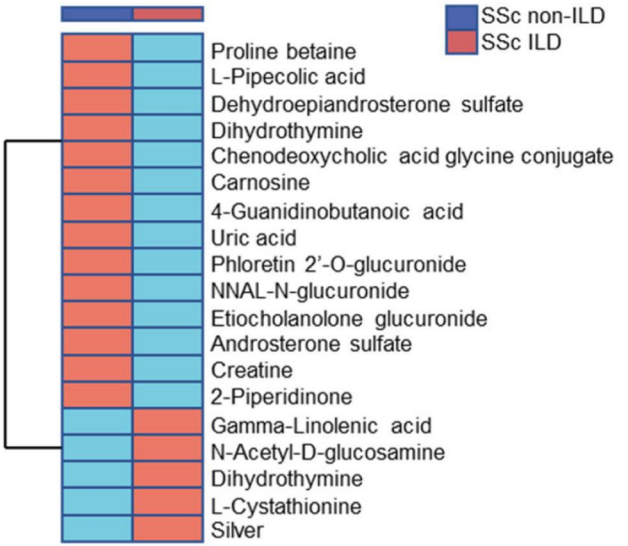


Allysine
L-Tryptophan
Urea
5-Acetylamino-6-formylamino-3-methyluracil

Figure 5







E

Machine learning models	Non_progressor	progressor
Linear SVM	69.4 [47.1-88.2]	56.2 [35.3-76.5]
RBF SVM	76.5 [76.5-76.5]	70.7 [52.9-88.2]
Random Forest	75.1 [64.7-82.4]	67.9 [52.9-82.4]
Nearest Neighbors	74.1 [58.8-82.4]	64.6 [47.1-82.4]
Gaussian Process	76.5 [76.5-76.5]	70.5 [52.9-88.2]
Decision Tree	62.5 [41.2-82.4]	57.2 [35.3-76.5]
Neural Net	68.0 [47.1-82.4]	57.5 [35.3-82.4]
AdaBoost	70.3 [52.9-82.4]	61.1 [41.2-82.4]
Gaussian Naive Bayes	60.3 [41.2-82.4]	52.6 [29.4-76.5]
QDA	66.9 [47.1-82.4]	52.0 [23.5-76.5]

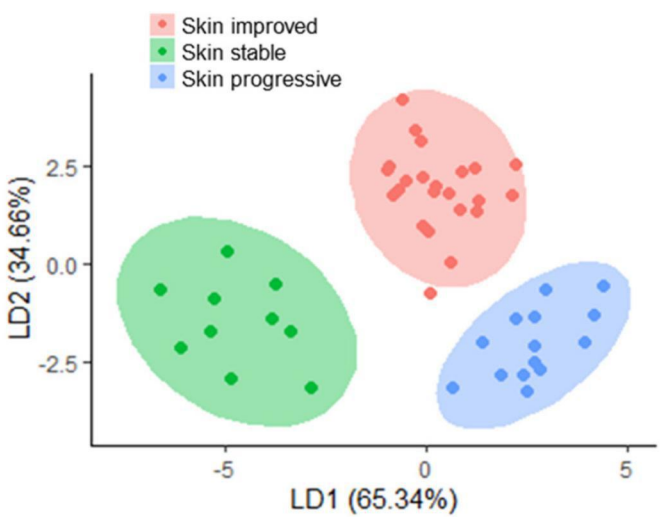


Figure 6